A-A-54475
30 August 1991

COMMERCIAL ITEM DESCRIPTION

REFRIGERATOR, MECHANICAL, BIOLOGICALS

(Blood Bank)

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a blood bank refrigerator.

Salient Characteristics:

The refrigerator shall be a 480 bag capacity (450 ml), reach-in, two door blood bank unit with temperature monitoring system and recording thermometer; removable, adjustable, slide-out shelves or drawers; triple-paned, cloudproof glass doors or stainless steel doors as specified by the processing activity; approximately 59" in width x 30" in depth x 77" in height.

The blood bank refrigerator, digital display type temperature monitoring system and recording thermometer shall be designed to maintain, monitor and record temperatures within the limits of the standard requirements established by the latest American Association of Blood Banks (AABB) Standard. Unit shall meet applicable requirements for refrigeration established by the American National Red Cross and the Food and Drug Administration.

The refrigerator shall be supplied with an appropriately sized compressor, condenser cooling fan, and interior circulation fan that all operate on the customer specified input voltage. The total operation of items mentioned above shall not consume more than 1800 watts. The refrigerator shall be supplied with line power fuse and motor overload protection.

AMSC N/A FSC 4110

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The temperature monitor shall constantly monitor the solution temperature as well as the upper/lower interior cabinet temperatures to ensure safe blood storage conditions.

Each cabinet shall be equipped with dual automatic or electronic controls set to maintain cabinet temperatures from 35°F to 39°F, forced air circulation for top to bottom temperature consistency, automatic condensate evaporator and automatic defrost system.

The refrigerator shall require sufficient integral insulation to allow for operation in the 35°F to 39°F range in ambient air temperature of up to 85°F without the necessity for continuous duty cycle or damage to the compressor.

Indicators shall include power "ON" and door "AJAR". The refrigerator shall have switches to control interior lights, battery test, and audible signal silencer and the ability to self-test alarm set points. The refrigerator shall have a keyed power on/off switch and a keyed alarm switch to prevent unauthorized shutdown and deactivation.

The unit shall be provided with a recording thermometer for recording blood temperatures over a 7-day period. The recording thermometer shall be a battery operated or spring-wound instrument with a continuous flow inking system that will provide up to 6 months of fine line temperature recording. The recording thermometer shall be furnished with 100 recording centigrade charts. Each chart shall be capable of providing a permanent record of each 24-hour period for 7 days. Device shall include the appropriate electrical connectors to add remote alarm devices at a future date at the user's option.

The refrigerator shall be equipped with stainless steel, adjustable, fully extendible drawers with perforated or solid bottoms or shelves. The cabinet interior shall be provided with sufficient illumination to enable inspection of contents and readability of temperature displays. The unit shall have casters (2 swivel with brakes, 2 rigid) and a moisture-tight chamber with drain, if applicable.

Electrical characteristics. The blood bank refrigerator shall operate over the full tolerance range of one or more of the following line voltage and frequency combinations as specified by the customer requisition.

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* 115 VAC plus or minus 10 percent, 50 Hz plus or minus 2 Hz (single-phase)

* 115 VAC plus or minus 10 percent, 60 Hz plus or minus 2 Hz (single-phase)

* 230 VAC plus or minus 10 percent, 50 Hz plus or minus 2 Hz (single-phase)

* 230 VAC plus or minus 10 percent, 60 Hz plus or minus 2 Hz (single-phase)

Change over devices, switches, circuits, or transformers if used for the purpose of voltage or frequency conversion, shall be clearly marked and identified in the appropriate documentation which accompanies the unit. Mechanical or electrical adjustments to effect the conversion shall not require the use of specialized tools or skills.

Identification marking. Each refrigeration unit shall bear the following data on one or more nameplates:

a. Name or registered trademark of the manufacturer.
b. Type or model number.
c. Serial number.
d. Electrical characteristics (in accordance with UL 471).
e. Refrigerant used, expressed in chemical and common form and amount used.
f. Factory test pressure.

Marking shall be permanent, distinct, and legible. Nameplates secured by adhesives shall not be used.

Finish. The refrigerator shall be fabricated from materials which are inherently resistant to corrosion or which are treated to render them corrosion resistant. Where painting or plating is used, appropriate cleaning and priming shall be performed prior to finishing. Finish shall provide a pleasing professional appearance and be free of gaps, scratches, chips and other imperfections.

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Underwriters Laboratories, Inc. The refrigerator shall conform to UL 471 Standard for Safety Refrigeration Units. Acceptable evidence of this shall be a UL listing mark or a certified test report from an independent laboratory acceptable to the Government.

Operators'and service data manuals. The contractor shall furnish manuals, handbooks and/or brochures containing complete operation, installation and service/maintenance instructions (including pictures or illustrations, as necessary) with complete schematics and wiring diagrams. All manuals, handbooks, and/or brochures shall be written in the English language and all schematics and wiring diagrams shall use American electrical and electronics symbols. The manuals shall include electrical data and connection diagrams for all applicable utilities. The instructions shall also contain a complete list of all replaceable parts showing part number, name and quantity required. All service notes, service memos, etc., pertaining to the equipment and issued subsequent to the printing of the original instructions shall also be included. These instructions shall be, as a minimum, that furnished to service engineers (or servicemen) that normally install and service the equipment for the company or distributor. When the system being procured includes, or will operate ancillary equipment, the service data shall include complete instructions and drawings which show interfacing of all system components.

Two (2) complete sets of the above information shall be packaged with each unit of the required equipment. Where multiple quantities of the same unit are to be delivered to the same ultimate consignee, then only a total of 2 service manuals need to be supplied to each consignee. However, 2 operational manuals still need to be supplied with each unit.

Workmanship. The mechanical, blood bank refrigerator shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One refrigerator, as specified, constitutes one unit.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of II. The unit of product for sampling purposes shall be one mechanical blood bank refrigerator.

Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of S-1. The unit of product for test purposes shall be one mechanical blood bank refrigerator, unless both 50 and 60 Hz units are ordered. One of each frequency shall be tested.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics. Tests shall include, but shall not be limited to, the following:

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Performance. The mechanical refrigerator blood bank shall be completely assembled in its operating configuration and tested to determine compliance with this specification. The refrigerator shall be operated to determine proper functioning. The refrigerator shall not be defective. Testing shall include the following:

Operation/Inspection:

1. Interior chamber drain.
2. Casters.
3. Moisture tight.
4. Seal and subsequent loss of cold air.
5. Door AJAR (The red LED flashes whenever the door is open or not completely closed).
6. Fail - Safe relay and external alarm adapter plug with "over/under-temp" alarm signal light.
7. Indicators and alarms.
8. Controls and switches.
9. Charts and ink.
10. Insulation.
11. Battery test switch.
12. Power failure indicator light.
13. Cabinet: Inspect the cabinet and parts thoroughly for any damage and missing parts. Inspect the interior/exterior of the refrigerator, including the mechanical equipment for special instructions and tags fastened at various points.
14. Line power fuse.
15. Motor overload protection.
16. Overall dimensions W X D X H

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17. Capacity.

18. Interior Lights.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

4110-01-304-7942

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project Number: 4110-0447

Location: REFRGCID.1BC

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